

 B. TECH. IN ELECTRICAL ENGINEERING (WORKING PROFESSIONAL)		SECOND YEAR SEM- III				
COURSE CODE	WPEE2004T	CREDITS ASSIGNED				
		THEORY	TUTORIAL	LAB	TOTAL HRS	TOTAL CREDITS
		3	-	-	3	-
COURSE TITLE	SENSORS & MEASUREMENTS	EVALUATION SCHEME				
		TA	MST	ESE	TOTAL MARKS	
		20	30	50	100	

COURSE OUTCOMES

Students should be able to:

1. Analyze different errors in measurement and their sources.
2. Apply different measuring principles AC as well as DC quantities.
3. Use correct measurement techniques for electrical and magnetic parameters
4. Understand and explain various transducers and their working
5. Understand and use CRO.

COURSE CONTENTS	
MODULE I	Fundamentals of Measurements
	Measurement System – Instrumentation – Classification and Characteristics of Transducers – Static and Dynamic – Errors in Measurements and their statistical analysis- methods of error analysis,- uncertainty analysis-expression of uncertainty: accuracy and precision index, propagation of errors– Calibration – Primary and secondary standards
MODULE II	Displacement, Pressure, Temperature Sensors
	Strain Gauge: Gauge factor, sensing elements, configuration, and unbounded strain gage. Capacitive transducer – various arrangements, Inductive transducer, LVDT, Passive types: RTD materials & range, relative resistance vs. temperature characteristics, thermistor characteristics, Active type: Thermocouple – characteristics.
MODULE III	Photoelectric and Piezo Electric Sensors
	Phototube, scintillation counter, photo multiplier tube (PMT), photovoltaic, photo conductive cells, photo diodes, phototransistor, comparison of photoelectric transducers. Optical displacement sensors and optical encoders. Piezoelectric active transducer- Equivalent circuit and its characteristics
MODULE IV	Signal Conditioning Circuits and Meters

	Functions of signal conditioning circuits, Preamplifiers, Concepts of passive filters, Impedance matching circuits, AC and DC Bridges – wheat stone bridge, Kelvin, Maxwell, Hay, Schering, Qmeter, PMMC, MI and dynamometer type instruments – DC potentiometer- Digital voltmeter – Multimeter.
MODULE V	Recording Devices and Advanced Sensors
	CRO – block diagram, CRT – vertical & horizontal deflection system, DSO, LCD monitor, PMMC writing systems, servo recorders, photographic recorder, magnetic tape recorder, Inkjet recorder, thermal recorder. Biosensors: transduction mechanism in a biosensor and Classification Electronic nose

TEXT BOOKS

1. “Electrical Measurements and Measuring Instruments”, by Golding. E. W, and Widdis F.C, 5th Edition, A. H. Wheeler & Company, 2003.
2. “A Course in Electrical and Electronics Measurements and Instrumentation”, by Sawhney A.K, 18th Edition, Dhanpat Rai & Company Private Limited, 2007.
3. “Measurement Systems - Application and Design”, Ernest O. Doebelin, TMH, 2007.

REFERENCE BOOKS

1. “Electronic Instrumentation”, by Kalsi. H. S, 2nd Edition, Tata McGraw Hill, 2004.
2. “Modern Electronic Instrumentation and Measurement Technique”, by Copper. W. D and Hlefrick A. D, 5th Edition, Prentice Hall of India, 2002.